

CSC 313: Formal Methods – Practice Question (Z Schemas)

A simple library system manages a collection of books. The system has the following requirements:

- Each book is uniquely identified. There is a given set **[Book]**.
- The library has a maximum capacity of **maxBooks = 500** (a constant).
- The state consists of:
 1. **stock** : the set of books currently on the shelves.
 2. **borrowed** : the set of books currently borrowed.
- Invariants:
 1. stock and borrowed are disjoint (no book is both on shelf and borrowed).
 2. The total number of books (in stock or borrowed) never exceeds maxBooks.

The system supports the following operations:

1. **AddBook** – adds a new book to the library (to stock).
 - Pre-condition: the book is not already in stock or borrowed, and the total number of books is less than maxBooks.
 - Post-condition: the book is added to stock, borrowed unchanged.
2. **BorrowBook** – a user borrows a book.
 - Pre-condition: the book is in stock.
 - Post-condition: the book is moved from stock to borrowed.
3. **ReturnBook** – a user returns a book.
 - Pre-condition: the book is in borrowed.
 - Post-condition: the book is moved from borrowed back to stock.
4. **IsAvailable** – a query operation that returns true if a given book is in stock, false otherwise. The state does not change.

Tasks (20 marks)

1. Axiomatic Definition (2 marks) - Define maxBooks = 500 as a natural number.
2. State schema Library (3 marks).
3. Initial state InitLibrary (3 marks).
4. Operation AddBook (3 marks).
5. Operation BorrowBook (3 marks).

6. Operation ReturnBook (3 marks).
7. Operation IsAvailable (3 marks) – use $\exists$ Library and output result! : bool.

Submission Instructions

- Write your answers clearly on the **official answer sheet** provided in class.
- Use **Z notation** as taught. You may refer to your notes and slides.
- Extra paper: If you need additional space, you may use regular paper. However, you must inform me early (before the deadline) if you do not have the official sheet.
- After completing, take a clear, well-lit photo or scan of every page.
- Upload the images in **a single PDF** to the course portal before **Sunday, 31 May, 11:59 PM**.
- **Important:** You **must** bring the original physical official answer sheet (and any extra sheets) to the next class after the deadline. Do not lose it.
 - **Reason:** This ensures the sheet is not kept for use in the final exam.
- Ensure your full name and matric number are written on every page.